CLASS 7 SCIENCE

Chapter-wise Revision Notes

(Based on NCERT 'Curiosity' Textbook of Science, Grade 7)

Covers all 12 chapters — key concepts, definitions, diagrams-in-words, and quick recap boxes

Chapter 1

1. The Ever-Evolving World of Science

This is an introductory chapter with no fixed factual content for exams. It sets the tone for the whole book and gives a preview of all the chapters ahead.

Key Ideas

- Science is not just a collection of facts — it is a process and a way of thinking based on curiosity, questioning, and exploration.
- Asking good questions is as important as finding answers. Scientists constantly ask: How do things work? Why do events happen? What patterns exist in nature?
- Different fields of science (physics, chemistry, biology, earth science) are interconnected — an idea in one area often helps explain another.
- The chapter previews the year's journey: properties of materials → electricity → metals/non-metals → physical & chemical changes → growing up (adolescence) → heat transfer → measurement of time and motion → life processes in animals and plants → light → Earth, Moon and Sun.
- Science involves responsibility too — understanding how human activity affects the natural world and society.

Quick Recap

- Science = a process of curiosity-driven questioning and exploration, not just facts.
- This chapter is a roadmap/preview of all topics covered in Grade 7.

Chapter 2

2. Exploring Substances: Acidic, Basic, and Neutral

2.1 Indicators — Testing the Nature of Substances

An indicator is a substance that shows a different colour in acidic and basic solutions, helping us identify the nature of a substance.

- **Litmus:** a natural dye obtained from lichens; available as blue and red litmus paper.
  - Acids turn blue litmus paper → red.
  - Bases turn red litmus paper → blue.
  - Neutral substances do not change the colour of either litmus paper.
- **Red rose extract:** turns red in acidic solutions and green in basic solutions.
- **Turmeric (haldi) paper:** turns red/reddish-brown in basic solutions; no change in acidic or neutral solutions (cannot distinguish acidic from neutral).
- **Olfactory indicators:** substances whose smell changes in acidic/basic medium, e.g., onion's smell fades when treated with an acid or base.

2.2 Properties of Acids and Bases

- Acids generally taste sour (e.g., citric acid in lemon/amla, tartaric acid in tamarind, ascorbic acid - Vitamin C, oxalic acid in vinegar/tomato).
- Bases generally taste bitter and feel soapy/slippery to touch (e.g., soap solution, baking soda solution, lime water, washing powder solution).
- Caution: never taste an unknown substance to test it.

2.3 Neutralisation

When an acid and a base are mixed in the right amounts, they cancel each other's effect — this is called neutralisation. It forms a salt and water, and heat is given out (exothermic).

- Acid + Base → Salt + Water + Heat

2.4 Neutralisation in Daily Life

- Ant/bee sting (acidic) → relieved by rubbing a mild base like baking soda.
- Acidic soil → treated by adding lime (a base).
- Basic soil → treated by adding organic matter/manure (which releases acids).
- Acidic factory waste → neutralised with basic substances before releasing into water bodies, to protect aquatic life.

Quick Recap

- Substances are acidic, basic, or neutral in nature.
- Acids turn blue litmus red; bases turn red litmus blue; neutral substances affect neither.
- Red rose extract: red in acid, green in base. Turmeric: turns red only in base.
- Acid + Base → Salt + Water + Heat (neutralisation).
- Neutralisation is used to treat insect stings, acidic/basic soil, and industrial waste.

Chapter 3

3. Electricity: Circuits and their Components

3.1 Components of a Simple Circuit

- **Electric cell:** a portable source of electrical energy; has a positive terminal (metal cap) and a negative terminal (flat metal disc).
- **Battery:** a combination of two or more cells, connected positive-to-negative in series, giving more energy/longer use.
- **Electric lamp (incandescent):** has a thin wire called a filament that glows white-hot when current passes through it, producing light.
- **LED (Light Emitting Diode):** has two terminals — a longer wire (positive) and a shorter wire (negative). Current flows through an LED in one direction only, so it glows only when connected the correct way round.
- **Switch:** a simple device that completes (closes) or breaks (opens) a circuit, allowing us to turn devices on/off.

3.2 The Electric Circuit

A complete, unbroken path for electric current to flow is called an electric circuit. Current flows only in a closed circuit (a circuit with no gaps).

- By convention, the direction of current in a circuit is taken from the positive terminal to the negative terminal of the cell (through the external circuit).
- **Circuit diagram:** a circuit drawn using standard symbols for cell, switch, bulb, wire, etc.

3.3 Conductors and Insulators

- **Conductors:** materials that allow electric current to pass through easily, e.g., metals like copper, iron, aluminium.
- **Insulators (poor conductors):** materials that do not allow current to pass through, e.g., rubber, plastic, wood, glass.

Quick Recap

- An electric cell has a positive (+) and negative (−) terminal; combining cells makes a battery.
- A filament lamp glows by heating a thin wire; an LED glows only when connected in the correct direction.
- A switch completes or breaks a circuit.
- Current flows from + to − terminal in a closed circuit (conventional direction).
- Conductors (e.g., metals) allow current to flow; insulators (e.g., rubber, plastic) do not.

Chapter 4

4. The World of Metals and Non-metals

4.1 Physical Properties of Metals

- **Lustre:** metals have a shine (lustrous); most non-metals are dull (non-lustrous).
- **Malleability:** metals can be beaten into thin sheets without breaking (e.g., gold, aluminium foil).
- **Ductility:** metals can be drawn into thin wires (e.g., copper wires).
- **Sonority:** metals produce a ringing sound when struck (e.g., bells, ghungroos); non-metals like coal or wood produce a dull sound.
- **Conduction of heat:** metals are generally good conductors of heat (used in cooking vessels); non-metals are poor conductors (used as handles).
- **Conduction of electricity:** metals are generally good conductors of electricity; non-metals are generally poor conductors (used as insulating handles for tools).

4.2 Effect of Air and Water on Metals

- **Corrosion:** damage to a metal surface caused by reaction with moisture/air over time.
- **Rusting:** corrosion of iron; iron reacts with oxygen and moisture in air to form a reddish-brown coating of iron oxide (rust).
- Different metals react differently with air and water — e.g., sodium reacts vigorously, while gold and silver barely react.

4.3 Chemical Properties: Oxides

- Metals react with oxygen to form metal oxides, which are generally basic in nature.
- Non-metals react with oxygen to form non-metal oxides, which are generally acidic in nature.
- Non-metals generally do not react with water.

4.4 Importance of Non-metals

Non-metals are essential too — e.g., oxygen for respiration, nitrogen for plant nutrition and proteins, carbon as the basis of all living matter.

Quick Recap

- Metals: lustrous, malleable, ductile, sonorous, and good conductors of heat & electricity.
- Non-metals generally lack these properties (exceptions exist).
- Metal oxides are basic; non-metal oxides are acidic.
- Rusting is the corrosion of iron caused by reaction with air and moisture.
- Both metals and non-metals are essential for everyday life and living organisms.

Chapter 5

5. Changes Around Us: Physical and Chemical

5.1 Physical Change

A change in which only the physical properties (size, shape, state) of a substance change, but no new substance is formed. Most physical changes are reversible.

- Examples: melting of ice, dissolving sugar in water, cutting paper, stretching a rubber band.

5.2 Chemical Change

A change in which one or more new substances with different properties are formed. It involves a chemical reaction, which can be shown using a chemical equation. Most chemical changes cannot be easily reversed.

- Examples: burning of paper, cooking food, rusting of iron, souring of milk.

5.3 Combustion and Rusting

- **Combustion:** a chemical change in which a combustible substance reacts with oxygen, releasing heat and/or light (burning).
- **Ignition temperature:** the lowest temperature at which a substance catches fire.
- **Rusting:** a slow chemical change where iron reacts with oxygen and moisture to form rust (iron oxide).

5.4 Reversible vs Irreversible, Desirable vs Undesirable

- Some changes can be reversed (e.g., melting and freezing of water); some cannot (e.g., burning of wood, cooking of an egg).
- Some changes are desirable (e.g., cooking food, ripening of fruit); some are undesirable (e.g., rusting of iron tools, souring of milk).

5.5 Slow Natural Changes

- **Weathering:** the breakdown of rocks into smaller pieces and eventually soil, due to the action of sun, water, wind, and living organisms; involves both physical changes (breaking) and chemical changes (reaction with air/water).
- **Erosion:** the removal/carrying away of weathered rock or soil particles by agents like flowing water and wind; this is a physical change.

Quick Recap

- Physical change: properties change but no new substance forms (often reversible).
- Chemical change: new substance(s) form, involves a chemical reaction (often irreversible).
- Combustion, cooking, and rusting are chemical changes.
- Ignition temperature: lowest temperature at which a substance catches fire.
- Changes can be reversible/irreversible and desirable/undesirable.
- Weathering breaks down rocks into soil; erosion carries away these particles (both shape the land slowly).

Chapter 6

6. Adolescence: A Stage of Growth and Change

6.1 What is Adolescence?

- Adolescence is the period of transition from childhood to adulthood, generally starting around age 10 and lasting until about 19 years.
- **Puberty:** the stage during adolescence when the body undergoes physical and biological changes, becoming capable of reproduction.
- Changes during adolescence are controlled mainly by chemical messengers in the body called hormones.

6.2 Reproductive and Physical Changes

- **Secondary sexual characteristics:** features that distinguish males from females but are not directly involved in reproduction (e.g., facial hair and deepening voice in boys; broader hips and breast development in girls).
- **Menstruation:** in girls, puberty also brings the start of the menstrual cycle — a discharge of blood roughly every 28–30 days, which continues until about 45–55 years of age (menopause).

6.3 Emotional and Behavioural Changes

- Adolescents often experience mood swings, increased sensitivity, desire for independence, and greater interest in peer relationships.
- These changes are a normal part of growing up and are best managed with support, guidance, and open communication with trusted adults.

6.4 Staying Healthy During Adolescence

- **Balanced diet:** nutritious food (proteins, vitamins, minerals, calcium, iron) supports the rapid growth of this stage.
- **Personal hygiene:** regular bathing and cleanliness become more important due to increased sweat and oil gland activity.
- **Physical activity:** regular exercise/sports keep the body and mind healthy.
- **Balanced social life:** maintaining good relationships with family and friends supports emotional well-being.
- **Avoiding addictive substances:** tobacco, alcohol, and drugs harm the body and mind — it is important to say no to them.

Quick Recap

- Adolescence: transition from childhood to adulthood (roughly age 10–19).
- Puberty brings physical, biological, and emotional changes controlled by hormones.
- Secondary sexual characteristics distinguish males and females.
- Menstruation begins at puberty in girls and continues until 45–55 years.
- Healthy diet, hygiene, exercise, good social life, and avoiding addictive substances help adolescents stay healthy.

Chapter 7

7. Heat Transfer in Nature

7.1 Conduction

Heat transfer from the hotter part of an object to a colder part, without the particles themselves changing position. Occurs mainly in solids.

- **Good conductors:** materials that let heat pass through easily, e.g., metals.
- **Poor conductors (insulators of heat):** materials that do not let heat pass easily, e.g., wood, plastic, air.

7.2 Convection

Heat transfer through the actual movement of particles. Occurs in liquids and gases (fluids).

- **Land and sea breeze:** during the day, land heats up faster than the sea; warm air over land rises and cooler sea air moves in (sea breeze). At night, the reverse happens, creating a land breeze.

7.3 Radiation

Heat transfer that does not require any medium — it can travel through empty space (vacuum). This is how heat from the Sun reaches the Earth.

- All objects continuously exchange heat with their surroundings through radiation.
- Conduction and convection need a medium; radiation does not.
- These principles are applied in designing houses (insulation) and choosing clothing (dark/light colours, fabric types) for different climates.

7.4 The Water Cycle

The continuous natural movement of water — upward as water vapour (evaporation) and downward as precipitation (rain/snow) — passing through air, soil, rocks, and plants, eventually returning to water bodies.

- **Infiltration:** the process by which surface water seeps into the soil and rocks below.
- This cycle is driven by heat from the Sun, linking heat transfer directly to weather and climate.

Quick Recap

- Three modes of heat transfer: conduction, convection, and radiation.
- Conduction: heat moves through a material without particle movement (mainly in solids).
- Convection: heat moves due to actual movement of particles (in liquids/gases); causes land and sea breeze.
- Radiation: heat transfer without any medium; how the Sun heats the Earth.
- Water cycle: evaporation → condensation → precipitation → infiltration, repeating continuously.

Chapter 8

8. Measurement of Time and Motion

8.1 Measuring Time

Early humans used repeating natural events (sunrise/sunset, Moon phases, seasons) to keep time, later inventing devices such as sundials (shadow position), water clocks (flow of water), hourglasses (flow of sand), and candle clocks (burning rate).

- **Simple pendulum:** a small bob attached to a string, free to swing; one complete to-and-fro swing is called an oscillation.
- **Time period:** the time taken to complete one oscillation; it stays constant for a pendulum of a given length at a given place (it does not depend on the mass of the bob or the angle of swing, for small swings).
- **SI unit of time:** second (symbol: s).

8.2 Speed

Speed tells us how fast or slow an object is moving.

- **Average speed = Total distance covered ÷ Total time taken**
- Common units: metres per second (m/s) and kilometres per hour (km/h).

8.3 Types of Motion

- **Uniform linear motion:** an object moving in a straight line covers equal distances in equal intervals of time (constant speed).
- **Non-uniform linear motion:** an object moving in a straight line covers unequal distances in equal intervals of time (changing speed).

Quick Recap

- Old time-measuring devices: sundial, water clock, hourglass, candle clock.
- Time period of a pendulum = time for one complete oscillation; constant for a given length at a place.
- SI unit of time is the second (s).
- Average speed = total distance ÷ total time.
- Uniform motion: constant speed; Non-uniform motion: changing speed.

Chapter 9

9. Life Processes in Animals

Life processes — nutrition, respiration, circulation, excretion, and reproduction — are essential activities that all living organisms must carry out to survive.

9.1 Nutrition (Digestion) in Animals

The human alimentary canal consists of: mouth → oesophagus → stomach → small intestine → large intestine → anus, along with associated organs — the liver and the pancreas — which aid digestion.

- Digested food is mainly absorbed through the walls of the small intestine into the blood.
- The large intestine absorbs remaining water and some salts from undigested food; the rest is expelled as faeces.
- **Ruminants:** grass-eating animals like cows and goats that partially chew and swallow food, later bring it back to the mouth to chew thoroughly ('chewing the cud'/rumination).

9.2 Respiration in Animals

- **Breathing:** a physical process — inhalation (taking air in) and exhalation (letting air out) through the lungs.
- Gas exchange (oxygen in, carbon dioxide out) occurs in tiny air sacs in the lungs called alveoli.
- **Respiration:** a chemical process in which cells use oxygen to break down glucose, releasing energy, carbon dioxide, and water.
- Breathing (physical) and respiration (chemical) are different processes, though related.
- Different animals have different breathing organs suited to their habitat — e.g., fish use gills (for water), insects use tracheae, earthworms breathe through their moist skin.

9.3 Circulation

- The circulatory system (heart + blood vessels) transports oxygen and nutrients to all body parts and carries away waste products.
- The heart pumps blood continuously through the body.

Quick Recap

- Life processes: nutrition, respiration, circulation, excretion, reproduction.
- Human digestive system: mouth → oesophagus → stomach → small intestine → large intestine → anus (+ liver, pancreas).
- Small intestine absorbs digested food; large intestine absorbs water/salts.
- Ruminants (cows, goats) chew the cud.
- Breathing = physical (inhalation/exhalation); Respiration = chemical (breakdown of glucose using O_2, releasing energy + CO_2 + water).
- Gas exchange occurs in alveoli of the lungs; different animals breathe differently (gills, tracheae, skin).

Chapter 10

10. Life Processes in Plants

10.1 How Plants Get Food: Photosynthesis

Leaves are the ‘food factories’ of a plant. Photosynthesis is the process by which green plants make their own food (glucose) using carbon dioxide and water, in the presence of sunlight and chlorophyll, also releasing oxygen.

- Word equation: Carbon dioxide + Water → (sunlight + chlorophyll) → Glucose + Oxygen
- **Stomata:** tiny pores on the surface of leaves (mostly on the underside) that allow exchange of gases (O_2 and CO_2) during photosynthesis and respiration, and also allow loss of water vapour (transpiration).
- **Chlorophyll:** the green pigment in leaves that captures sunlight energy needed for photosynthesis.

10.2 Transport in Plants

- **Xylem:** transports water and minerals upward from the roots to all parts of the plant.
- **Phloem:** transports prepared food (made in leaves) to all parts of the plant, including for storage.

10.3 Respiration in Plants

Plants also respire — they break down glucose using oxygen to release energy, producing carbon dioxide and water, just like animals. This happens all the time, in all parts of the plant, day and night.

Photosynthesis vs Respiration (Comparison)

Feature	Photosynthesis	Respiration
Raw materials	CO_2 + Water	Glucose + O_2
Products	Glucose + O_2	CO_2 + Water + Energy
Requires sunlight?	Yes (only in daytime)	No (occurs all the time)
Occurs in	Mainly in green/leaf cells (chlorophyll)	All living cells of the plant

Quick Recap

- Photosynthesis: CO_2 + Water → (sunlight + chlorophyll) → Glucose + Oxygen.
- Leaves are the food factories of a plant; stomata allow gas exchange.
- Xylem transports water/minerals (upward); phloem transports food (to all parts).
- Plants also respire — breaking down glucose with oxygen to release energy, CO_2, and water.

Chapter 11

11. Light: Shadows and Reflections

11.1 Sources of Light

- **Luminous objects:** objects that emit their own light, e.g., the Sun, a burning candle, an electric bulb.
- Non-luminous objects (e.g., the Moon) only reflect light from a luminous source; they do not produce their own light.

11.2 Light Travels in a Straight Line

Light travels in straight lines, which is why it can be blocked by objects, creating shadows, and is why we cannot see around corners without help.

11.3 Transparent, Translucent, and Opaque Materials

- **Transparent:** allow light to pass through almost completely (e.g., clear glass, water).
- **Translucent:** allow light to pass through only partially (e.g., frosted glass, butter paper).
- **Opaque:** do not allow light to pass through at all (e.g., wood, metal, cardboard).

11.4 Shadows

A shadow forms when an opaque or translucent object blocks light travelling in a straight line.

- Opaque objects form dark, well-defined shadows; translucent objects form lighter (partial) shadows; some transparent objects may form faint shadows.

11.5 Reflection of Light

The bouncing back of light when it strikes a surface (such as a mirror) is called reflection.

11.6 Image in a Plane Mirror

The image formed by a plane mirror is:

- Same size as the object
- Erect (upright, not inverted)
- Virtual (cannot be obtained on a screen)
- **Laterally inverted** (left appears as right and vice versa).

11.7 Pinhole Camera

A simple device with a tiny hole that lets light in and forms an inverted (upside-down) image of an object on a screen placed inside it — this demonstrates that light travels in straight lines.

11.8 Useful Applications

- **Periscope:** uses two plane mirrors to let us see objects around corners or above an obstacle (used in submarines).
- **Kaleidoscope:** uses mirrors arranged at an angle with small coloured objects to create beautiful repeating patterns.

Quick Recap

- Luminous objects emit their own light; light travels in a straight line.
- Materials are transparent, translucent, or opaque based on how much light passes through.
- A shadow forms when light is blocked; opaque objects make the darkest shadows.
- Plane mirror image: same size, erect, virtual, and laterally inverted.
- Pinhole camera forms an inverted image, proving light travels straight.
- Periscope and kaleidoscope are practical applications of reflection using mirrors.

Chapter 12

12. Earth, Moon, and the Sun

12.1 Rotation of the Earth

The Earth spins on its own imaginary axis, completing one full rotation in about 24 hours.

- Rotation (West to East) causes day and night, and also the apparent daily motion of the Sun, Moon, and stars across our sky (rising in the east, setting in the west).

12.2 Revolution of the Earth

The Earth moves around the Sun in a roughly circular path called its orbit, completing one revolution in about 365 days (1 year).

- **Tilted axis:** the Earth's axis of rotation is not perpendicular to its orbital plane — it is tilted.
- **Seasons:** caused by the combination of the Earth's tilted axis and its revolution around the Sun — different parts of the Earth receive different amounts of sunlight at different times of the year.
- The view of the night sky (visible stars and constellations) changes through the year as the Earth revolves around the Sun.

12.3 Eclipses

- **Solar eclipse:** occurs when the Moon comes between the Sun and the Earth, blocking sunlight from reaching parts of the Earth (Sun → Moon → Earth in line). Happens on a new moon day.
- **Lunar eclipse:** occurs when the Earth comes between the Sun and the Moon, blocking sunlight from reaching the Moon (Sun → Earth → Moon in line). Happens on a full moon day.

Quick Recap

- Rotation: Earth spins on its axis once in ~24 hours → causes day and night.
- Revolution: Earth orbits the Sun once in ~365 days → along with the tilted axis, causes seasons.
- Solar eclipse: Moon comes between Sun and Earth.
- Lunar eclipse: Earth comes between Sun and Moon.